Classical Profiled-Attack Baseline

To provide a classical side-channel baseline for the machine-learning experiments, we evaluated a Multivariate Gaussian template (GT) attack [1,2] on a representative Stage-I single-pair configuration. The baseline uses the same held-out attack corpus, target byte, region of interest, and differential representation as the corresponding CNN experiment. This is because GT in its classical sense would be futile we have adapted it to the differential representation which needs not "plaintext"-hypothesis as classical model-base/template attacks.

Specifically, we use the multiplicative pair representation d_{mul}, targeting Byte 1 of the pair delta (zero-based array index 0), with samples $\{80, \ldots 144\}$ as the region of interest. Each attack example consists of two *primitive leakages* forming one paired execution. Consequently, an attack budget of N examples corresponds to $2N$ *primitive leakages*.

The template model is trained only on the profiling corpus. Points of interest are selected using the profiling-set SNR, after which the ten highest-ranking sufficiently separated samples are retained. A Multivariate Gaussian template is fitted for each of the 256 possible delta-byte values using a class-specific mean and a shared pooled full covariance matrix. A small ridge regularization term of 10^{-3} is applied to the covariance estimate. The attack set is not used for POI selection or template fitting.

For an attack observation x, the template attack computes a log-likelihood score for each candidate $\delta \in \{0, \ldots, 255\}$. Scores are accumulated across attack examples in the log domain, and the 256 candidates are ranked according to their accumulated likelihood. Guessing Entropy (GE) and Success Rates SR_1, SR_2, and SR_4 are evaluated over 170 randomized attack subsets for each trace budget, using the same evaluation convention as the neural-network experiments.

Table 1: Gaussian template-attack baseline on the Stage-I d_{mul} single-pair setting. Each attack example contains two *primitive leakages*.

N	**Primitive leakages**	**GE**	$\textbf{SR}_1$ (%)	$\textbf{SR}_2$ (%)	$\textbf{SR}_4$ (%)
1	2	125.00	0.00	0.00	0.00
5	10	113.02	0.00	0.00	0.00
10	20	105.30	0.00	0.00	0.00
20	40	98.85	0.00	0.00	0.59
50	100	81.05	0.00	0.00	0.00
100	200	70.56	0.00	0.00	0.59
200	400	63.92	0.00	0.00	0.00
500	1000	52.08	0.00	0.00	0.00
1000	2000	48.98	0.00	0.00	0.00
2000	4000	53.00	0.00	0.00	0.00

As shown in Table 1, the classical multivariate template attack extracts measurable information from the same differential representation: the mean correct-candidate rank decreases from a GE of 125.00 with one attack example to 48.98 with 1,000 attack examples. However, this improvement does not translate into reliable top-k recovery. Across the evaluated range of up to 2,000 attack examples (4,000 *primitive leakages*), SR_1 and SR_2 remain at 0%, while SR_4 is effectively zero, reaching only 0.59% at isolated budgets. At the maximum evaluated budget, the mean correct-candidate rank remains 53.00.

In contrast, the CNN evaluated on the same Stage-I d_{mul} representation achieves rank-1 recovery at a substantially smaller attack budget. This comparison shows that the differential representation alone does not account for the reported attack performance: the classical profiled model extracts some information from the same representation, but does not achieve reliable recovery within the evaluated budget, whereas the CNN exploits the available leakage considerably more effectively.

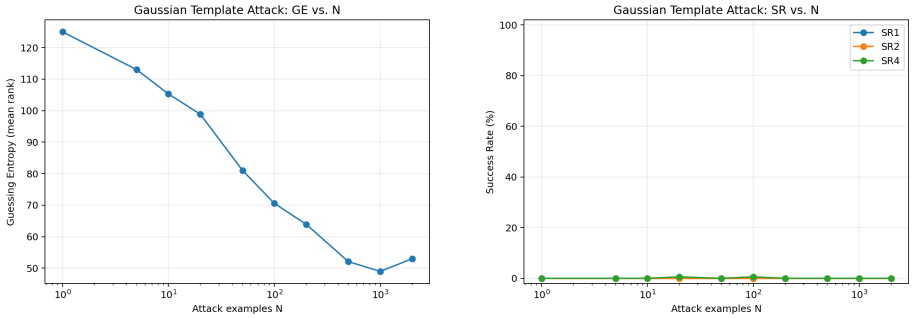


Fig. 1: Classical Gaussian template-attack baseline on the Stage-I d_{mul} representation. Left: mean correct-candidate rank (GE) as a function of the number of attack examples. Right: top-k success rates. One attack example contains two *primitive leakages*.

References

1. S. Chari, J. R. Rao, and P. Rohatgi. Template attacks. In *Cryptographic Hardware and Embedded Systems – CHES 2002*, volume 2523 of *Lecture Notes in Computer Science*, pages 13–28. Springer, 2002.
2. O. Choudary and M. G. Kuhn. Efficient template attacks. In *Smart Card Research and Advanced Applications – CARDIS 2013*, volume 8419 of *Lecture Notes in Computer Science*, pages 253–270. Springer, 2014.